

Predictive Flood Intelligence for Urban Disaster Response

A spatiotemporal system that learns how flooding spreads through a city's streets, and listens for who's asking for help — in the language they're actually using.

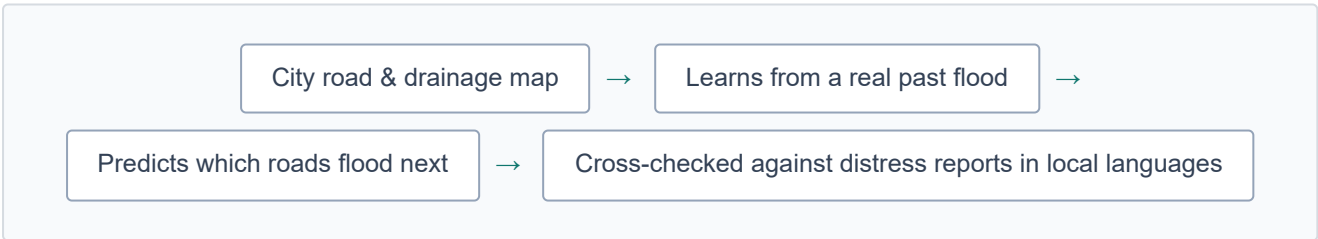
The Problem

When a city floods, the response rarely matches how flooding actually behaves. Warning systems issue blanket alerts across an entire ward or city, even though water spreads through a road network unevenly — block by block, shaped by drainage, elevation, and whatever's already flooded next door.

Meanwhile, when people report being stranded, most crisis-response tools are trained on English social media, missing the code-mixed regional-language posts that are how people in Indian cities actually communicate in an emergency. The result is a mismatch on both ends: warnings too vague to act on, and distress calls that go unheard because they weren't posted in English.

What We're Proposing

We propose a system that treats flood risk as what it actually is — a process that spreads through a connected network over time — and treats crisis reporting as a multilingual, location-grounded problem rather than an English-only one. Built and validated against a real, well-documented flood, it's designed to answer a question current systems can't: *given how it's flooding right now, which street floods next, and who there needs help?*



How We'll Build It

1 · Flood Propagation Model

We convert the road network into a graph where each street segment is a node and shared intersections form the connections — so the model reasons about how flooding on one street affects its neighbors, not just how much rain fell on it. We train a temporal graph neural network (built on PyTorch Geometric Temporal) on the 2015 Chennai floods, using satellite imagery, rainfall records, and news reports fused together as ground truth.

To prove the model earns its complexity, we benchmark it head-to-head against a simple rule-based baseline ("if it rains this much, this road floods"). The claim is falsifiable: the learned model should catch cascading flood spread that the static rule misses.

2 · Multilingual Distress Detection

We fine-tune a pretrained multilingual language model (IndicBERT / MuRIL) to flag genuine distress in code-mixed social media posts — the mix of English, Hindi, Kannada, and Tamil people actually text in during an emergency.

Because place names usually survive language-switching intact, we resolve each flagged post to a real coordinate by matching it against a gazetteer built directly from the same road network used in the flood model — no separate location dataset required.

Why It Matters

REACHES PAST THE ENGLISH INTERNET

Disaster-response tools are almost all built and tested on English social media, quietly deprioritizing exactly the communities most likely to be under-resourced during a flood. We're building for code-mixed regional language posts from day one, not as an afterthought.

DOESN'T JUST CHASE THE LOUDEST SIGNAL

Social media over-represents people who are online, connected, and vocal. A response system built purely on that signal quietly under-serves quieter, higher-vulnerability neighborhoods — the ones who often need help the most.

WARNINGS THAT MEAN SOMETHING

"Your ward may flood" tells a family nothing they can act on. "This road, in the next few hours" is something a person can act on, and something a responder can use to decide where to send help first.

GROUNDNED, NOT SPECULATIVE

This isn't modeled on invented data. It's trained against a real, well-documented flood — Chennai, 2015 — using an official satellite-derived flood map and a methodology already peer-reviewed for that exact event.

What We're Committing To

Ward-scale flood propagation model — the predictive core of the system

Core commitment

Multilingual distress detection & location matching

Core commitment

Resources We'll Draw On

Real-World Data

- OpenStreetMap road & drainage network for the study wards
- SRTM 30m elevation model, for slope and low-lying terrain
- Open-Meteo / IMD hourly rainfall records
- Sentinel-1 satellite radar and NRSC Bhuvan's official flood footprint for the 2015 Chennai floods
- News & traffic-advisory archives from the event
- Census 2011 ward demographics (*optional layer*)

Tools & Compute

- PyTorch Geometric Temporal — pre-built temporal graph-neural-network layers
- HuggingFace Transformers — IndicBERT / MuRIL for multilingual text
- geopandas, networkx, rapidfuzz — geospatial processing & matching
- Google Earth Engine — satellite imagery processing
- Google Colab / Kaggle free-tier GPUs — training compute

The two core pieces are deliberately built to plug into each other — the same neighborhood map that trains the flood model doubles as the location reference for the distress detector — so the result reads as one connected system, grounded in one real place and one real flood, rather than separate experiments bolted together.